of 0.4 percent of GDP. From 2020 to 2024, the baseline assumes an overall fiscal deficit that starts at 2.4 percent of GDP and decreases gradually to 2.2 percent of GDP. The first alternative scenario assumes a *temporary* relaxation (for 6 years) of fiscal policy with an average overall deficit around 2.7 percent of GDP and a maximum deficit of 3.0 percent. The second alternative scenario assumes a *permanent* relaxation of the overall fiscal deficit, reaching on average 2.3 percent of GDP from 2019 to 2075.

5. Fiscal position remains sustainable in the long-run under all three scenarios, while the debt ratio is higher under the permanent relaxation. In the baseline scenario, the debt-to-GDP ratio stabilizes at a level of 27.0 percent of GDP in the medium and long term; and the debt-to-revenue ratio remains at around 249 percent. In the scenario of temporary relaxation, the debt-to-GDP ratio rises slightly in the short term and stabilizes at 29.5 percent of GDP in the long term while debt-to-revenue ratio increases to 255 percent. Under the permanent relaxation scenario, the debt ratio rises until 2035 and stabilizes at 32.0 percent of GDP, with the debt-to-revenue ratio reaching 269 percent. Under all three scenarios the debt-to-GDP does not exceed an indicative benchmark for countries with market access at 60 percent.

6. Sensitivity analysis suggests that Guatemala's debt burden indicators are quite resilient to macroeconomic shocks. Five sensitivity tests are considered, including a shock to the primary balance, a shock to the real GDP growth, a shock to the real interest rate, a shock to the real exchange rate as well as a shock combining all the above. The size of each shock was based on the historical standard deviations of the corresponding variables.

- *Real GDP Growth Shock.* GDP growth rate is reduced by 1 standard deviation for 2 consecutive years, dropping growth rate to 2.5 and 2.7 percent in 2020 and 2021, respectively. Decline in growth leads to lower inflation (0.25 percentage points per 1 percentage point decrease in GDP growth).

- *Primary Surplus Shock.* Shock is equivalent 0.3 percent of GDP (half of the 10-year historical standard deviation). Shock triggers an increase in interest rates of 25 basis points for every percentage point of GDP worsening in the primary balance.

- *Interest Rate Shock.* Interest rate increases by 200 basis points respect to the baseline.

- *Real Exchange Rate Shock.* Shock translates to a nominal exchange rate depreciation of 10 percent; pass-through to inflation with an elasticity of 0.25.

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Figure III.1. Guatemala: Public Sector Debt Sustainability Analysis (DSA) – Baseline Scenario

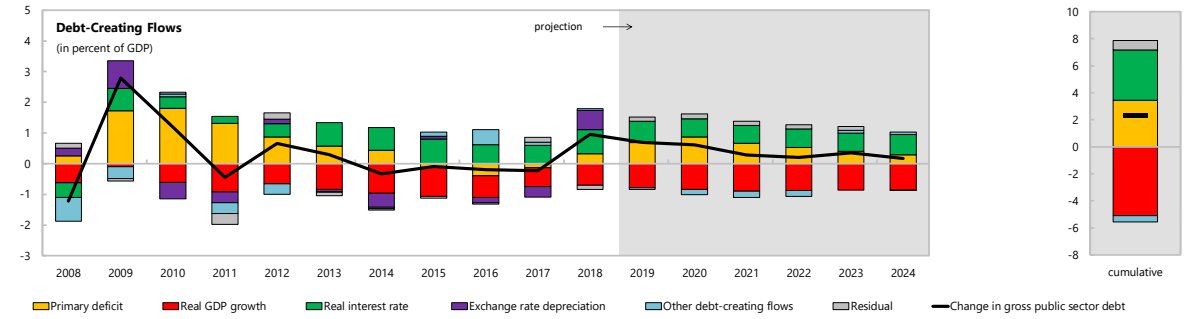
(In percent of GDP unless otherwise indicated)

Debt, Economic and Market Indicators 1/

	Actual			Projections						As of February 28, 2019		
	2008-2016 2/	2017	2018	2019	2020	2021	2022	2023	2024	Sovereign Spreads		
Nominal gross public debt	23.6	23.8	24.7	25.4	26.0	26.3	26.5	26.9	27.0	EMBIG (bp) 3/		213
Public gross financing needs	3.5	2.3	2.7	3.3	3.7	3.5	3.4	3.0	3.2	5Y CDS (bp)		N/A
Real GDP growth (in percent)	3.2	2.8	3.1	3.4	3.5	3.7	3.6	3.5	3.5	Ratings	Foreign	Local
Inflation (GDP deflator, in percent)	4.6	3.6	2.8	3.4	4.0	4.3	4.4	4.7	4.7	Moody's	Ba1	Ba1
Nominal GDP growth (in percent)	8.0	6.5	6.0	6.8	7.7	8.2	8.1	8.3	8.3	S&Ps	BB-	BB
Effective interest rate (in percent) 4/	6.9	6.4	6.4	6.5	6.7	6.8	7.0	7.2	7.5	Fitch	BB	BB

Contribution to Changes in Public Debt

	Actual			Projections							
	2008-2016	2017	2018	2019	2020	2021	2022	2023	2024	cumulative	debt-stabilizing primary balance 9/
Change in gross public sector debt	0.3	-0.2	1.0	0.7	0.6	0.3	0.2	0.3	0.2	2.3	
Identified debt-creating flows	0.3	-0.4	1.1	0.6	0.5	0.1	0.1	0.2	0.2	1.6	
Primary deficit	0.7	-0.1	0.3	0.7	0.9	0.7	0.5	0.4	0.3	3.5	-0.1
Primary (noninterest) revenue and grants	11.4	10.8	10.6	10.4	10.5	10.6	10.7	10.8	10.9	63.8	
Primary (noninterest) expenditure	12.1	10.7	10.9	11.1	11.3	11.2	11.2	11.2	11.2	67.2	
Automatic debt dynamics 5/	-0.3	-0.4	0.7	-0.1	-0.2	-0.3	-0.3	-0.3	-0.2	-1.4	
Interest rate/growth differential 6/	-0.2	0.0	0.1	-0.1	-0.2	-0.3	-0.3	-0.3	-0.2	-1.4	
Of which: real interest rate	0.5	0.6	0.8	0.7	0.6	0.6	0.6	0.6	0.7	3.7	
Of which: real GDP growth	-0.7	-0.6	-0.7	-0.8	-0.8	-0.9	-0.9	-0.9	-0.9	-5.1	
Exchange rate depreciation 7/	0.0	-0.3	0.6	...	...	...	...	...	...	...	
Other identified debt-creating flows	-0.1	0.1	0.1	-0.1	-0.2	-0.2	-0.2	0.1	0.1	-0.5	
Use of deposits	-0.1	0.1	0.1	-0.1	-0.2	-0.2	-0.2	0.1	0.1	-0.5	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Privatization receipts (+ reduces financing needs)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes 8/	0.0	0.2	-0.1	0.1	0.2	0.1	0.1	0.1	0.0	0.7	



Source: IMF staff.
1/ Public sector is defined as central government.
2/ Based on available data.
3/ EMBIG.
4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.
5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).
6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.
7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).
8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.
9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

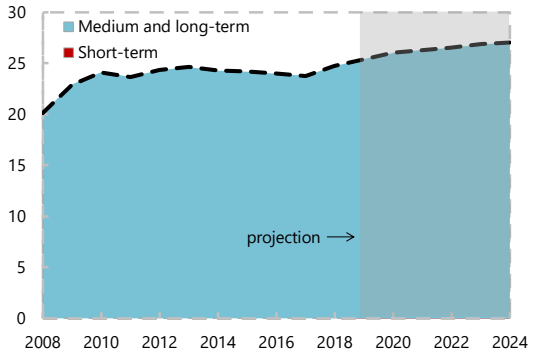
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Figure III.2. Guatemala: Public DSA – Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

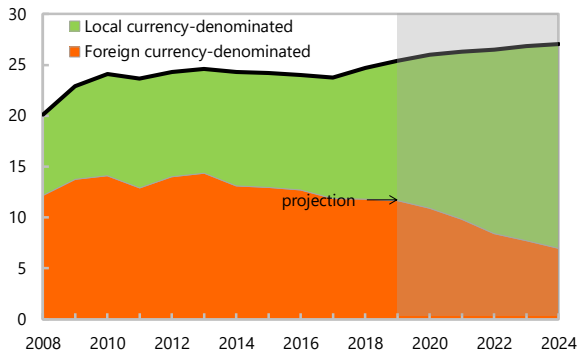
By Maturity

(In percent of GDP)

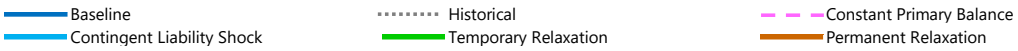


By Currency

(In percent of GDP)

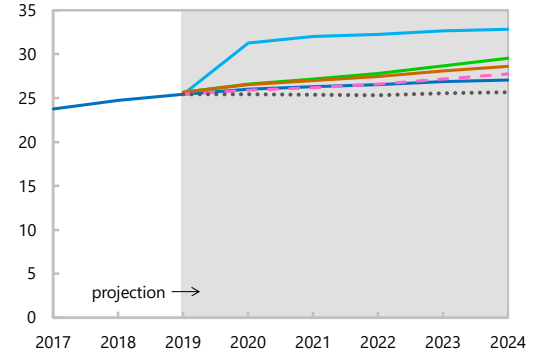


Alternative Scenarios



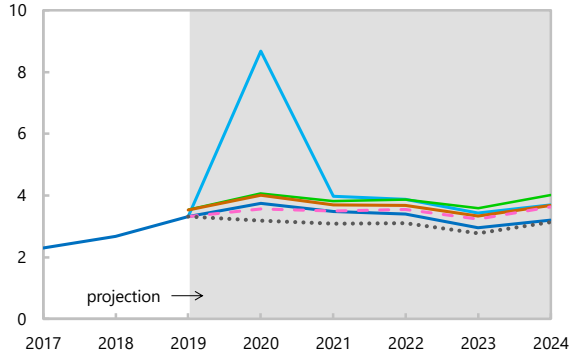
Gross Nominal Public Debt

(In percent of GDP)



Public Gross Financing Needs

(In percent of GDP)



Underlying Assumptions

(In percent)

Baseline Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	3.4	3.5	3.7	3.6	3.5	3.5
Inflation	3.4	4.0	4.3	4.4	4.7	4.7
Primary Balance	-0.7	-0.9	-0.7	-0.5	-0.4	-0.3
Effective interest rate	6.5	6.7	6.8	7.0	7.2	7.5
Constant Primary Balance Scenario						
Real GDP growth	3.4	3.5	3.7	3.6	3.5	3.5
Inflation	3.4	4.0	4.3	4.4	4.7	4.7
Primary Balance	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Effective interest rate	6.5	6.7	6.8	7.0	7.2	7.5
Temporary Relaxation						
Real GDP growth	3.4	3.5	3.7	3.6	3.5	3.5
Inflation	3.4	4.0	4.3	4.4	4.7	4.7
Primary Balance	-0.9	-1.2	-1.0	-0.9	-0.9	-0.9
Effective interest rate	6.5	6.7	6.9	7.1	7.3	7.6

Historical Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	3.4	3.8	3.8	3.8	3.8	3.8
Inflation	3.4	4.0	4.3	4.4	4.7	4.7
Primary Balance	-0.7	-0.3	-0.3	-0.3	-0.3	-0.3
Effective interest rate	6.5	6.7	6.8	7.1	7.2	7.6
Contingent Liability Shock						
Real GDP growth	3.4	2.5	2.7	3.6	3.5	3.5
Inflation	3.4	3.7	4.0	4.4	4.7	4.7
Primary Balance	-0.7	-5.4	-0.7	-0.5	-0.4	-0.3
Effective interest rate	6.5	8.1	7.2	7.3	7.5	7.7
Permanent Relaxation						
Real GDP growth	3.4	3.5	3.7	3.6	3.5	3.5
Inflation	3.4	4.0	4.3	4.4	4.7	4.7
Primary Balance	-0.9	-1.1	-0.8	-0.8	-0.7	-0.6
Effective interest rate	6.5	6.7	6.9	7.1	7.3	7.6

Source: IMF staff.

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